

NEET 2024

```
In [6]: import numpy as np
import pandas as pd

dataset = pd.read_csv("all_data.csv")
dataset.dropna()
dataset.info
```

```
Out[6]: <bound method DataFrame.info of
0      272807  KASTURBA GIRLS PU COLLEGE, KASTURBA GIRLS PU C...
1      272807  KASTURBA GIRLS PU COLLEGE, KASTURBA GIRLS PU C...
2      272807  KASTURBA GIRLS PU COLLEGE, KASTURBA GIRLS PU C...
3      272807  KASTURBA GIRLS PU COLLEGE, KASTURBA GIRLS PU C...
4      272807  KASTURBA GIRLS PU COLLEGE, KASTURBA GIRLS PU C...
...      ...
2333157  250301  AYESHA ALI ACADEMY, KANIPORA KULGAM, J&K, ANAN...
2333158  250301  AYESHA ALI ACADEMY, KANIPORA KULGAM, J&K, ANAN...
2333159  250301  AYESHA ALI ACADEMY, KANIPORA KULGAM, J&K, ANAN...
2333160  250301  AYESHA ALI ACADEMY, KANIPORA KULGAM, J&K, ANAN...
2333161  250301  AYESHA ALI ACADEMY, KANIPORA KULGAM, J&K, ANAN...

      center_city  center_state  serial_no  score
0  SHIVAMOGA (SHIMOGA)  KARNATAKA         1     42
1  SHIVAMOGA (SHIMOGA)  KARNATAKA         2    172
2  SHIVAMOGA (SHIMOGA)  KARNATAKA         3     42
3  SHIVAMOGA (SHIMOGA)  KARNATAKA         4    222
4  SHIVAMOGA (SHIMOGA)  KARNATAKA         5    129
...      ...      ...      ...      ...
2333157  ANANTNAG  JAMMU & KASHMIR     555    137
2333158  ANANTNAG  JAMMU & KASHMIR     556     83
2333159  ANANTNAG  JAMMU & KASHMIR     557    513
2333160  ANANTNAG  JAMMU & KASHMIR     558    135
2333161  ANANTNAG  JAMMU & KASHMIR     559    118

[2333162 rows x 6 columns]>
```

```
In [44]: dataset.describe().round(3)
```

```
Out[44]:
```

	center_id	serial_no	score	Rank
count	2333162.000	2333162.000	2333162.000	2333162.000
mean	322402.048	290.234	217.157	1163725.674
std	104959.799	217.144	166.240	672164.038
min	110101.000	1.000	-180.000	1.000
25%	260202.000	124.000	87.000	581722.000
50%	311119.000	251.000	163.000	1165724.000
75%	412206.000	408.000	313.000	1749236.000
max	991401.000	1968.000	720.000	2333161.000

acc to shiksha.com the cutoffs for neet 2024 were calculated based on

1. For Unreserved/General - 50th percentile of marks meaning the cutoff range is 720 - 164
2. For SC/ST/OBC - 40th percentile of marks meaning the cutoff range is 161 - 127

```
In [11]: # finding cutoff based on given score from the data
scores = dataset["score"]

general = np.percentile(scores, 50)
reserved = np.percentile(scores, 40)

print("General cutoff 2024: ", general)
print("Reserved cutoff 2024: ", reserved)
```

```
General cutoff 2024: 163.0
Reserved cutoff 2024: 128.0
```

```
In [19]: # calculating Ranks using the dataset

dataset['Rank'] = dataset["score"].rank(ascending=False, method='min').astype(int)
dataset = dataset.sort_values(by='Rank').reset_index(drop=True)
print("Top 15 ranked")
```

```
dataset.head(15)
```

Top 15 ranked

Out[19]:		center_id	center_name	center_city	center_state	serial_no	score	Rank
0	410801	PAAVAI ENGINEERING COLLEGE, NH-44 PACHAL (PO) ...		NAMAKKAL	TAMIL NADU	414	720	1
1	311120	MODERN SCHOOL, OPP SUNDER BISCUIT FACTORY 8TH ...		NAGPUR	MAHARASHTRA	460	720	1
2	390601	ST. ANTHONYS SR. SEC. SCHOOL, GOVERDHAN VILAS ...		UDAIPUR	RAJASTHAN	459	720	1
3	220106	SHREE NARAYANA CENTRAL SCHOOL, OPP ISRO JODHPU...		AHMEDABAD	GUJARAT	458	720	1
4	311250	MATOSHRI PRATISHTHAN, VISHWABHARATI POLYTECHNI...		NANDED	MAHARASHTRA	147	720	1
5	271216	NAGARJUNA VIDYANIKETAN, #104 IVRI ROAD BEHIND ...		BENGALURU- URBAN	KARNATAKA	158	720	1
6	390527	DECENT SCHOOL, DEOLI ARAB-RAIPURA ROAD- RAJNAGA...		KOTA	RAJASTHAN	165	720	1
7	280826	SAPPHIRE CENTRAL SCHOOL, PEACE VALLEY CAMPUS M...		KOZHIKODE/CALICUT	KERALA	43	720	1
8	390301	ALPHA INTERNATIONAL ACADEMY, KATARIA FARM, VIS...		JAIPUR	RAJASTHAN	691	720	1
9	392344	SARASWATI PUBLIC SCHOOL, BEHIND KAMALA MODI MA...		SIKAR	RAJASTHAN	212	720	1
10	220804	J B DIAMONDS &KARP IMPEXVIDYA SANKUL, OPP. DIA...		SURAT	GUJARAT	177	720	1
11	460902	D.A.V SCHOOL, VILLAGE: JOGIVITA FULBARI P.O MA...		SILIGURI	WEST BENGAL	58	720	1
12	410107	ST.PETERS INSTITUTE OF HIGHER EDUCATION AND RE...		CHENNAI	TAMIL NADU	332	720	1
13	120504	DHANEKULA INSTITUTE OF ENGINEERING & TECHNOLOG...		VIJAYAWADA	ANDHRA PRADESH	383	720	1

	center_id	center_name	center_city	center_state	serial_no	score	Rank
14	270715	SHRI DHARMASTHALA MANJUNATHESHWARA SCHOOL, SDM...	MANGALURU (MANGALORE)	KARNATAKA	70	720	1

checking eligibility for top Medical Colleges, closing ranks

1. AIIMS Delhi - 47
2. AIIMS Bhopal - 79
3. AIIMS Rishikesh - 594
4. AIIMS Bhubaneshwar - 655
5. AIIMS Nagpur - 953

```
In [ ]: dataset['AIIMS Delhi'] = dataset['Rank'].apply(lambda x: 'Eligible' if x < 47 else 'Not Eligible')
dataset['AIIMS Bhopal'] = dataset['Rank'].apply(lambda x: 'Eligible' if x < 79 else 'Not Eligible')
dataset['AIIMS Rishikesh'] = dataset['Rank'].apply(lambda x: 'Eligible' if x < 594 else 'Not Eligible')
dataset['AIIMS Bhubaneshwar'] = dataset['Rank'].apply(lambda x: 'Eligible' if x < 655 else 'Not Eligible')
dataset['AIIMS Nagpur'] = dataset['Rank'].apply(lambda x: 'Eligible' if x < 953 else 'Not Eligible')

dataset.iloc[150:350]
```

```
In [42]: dataset.iloc[950:965]
```

Out[42]:

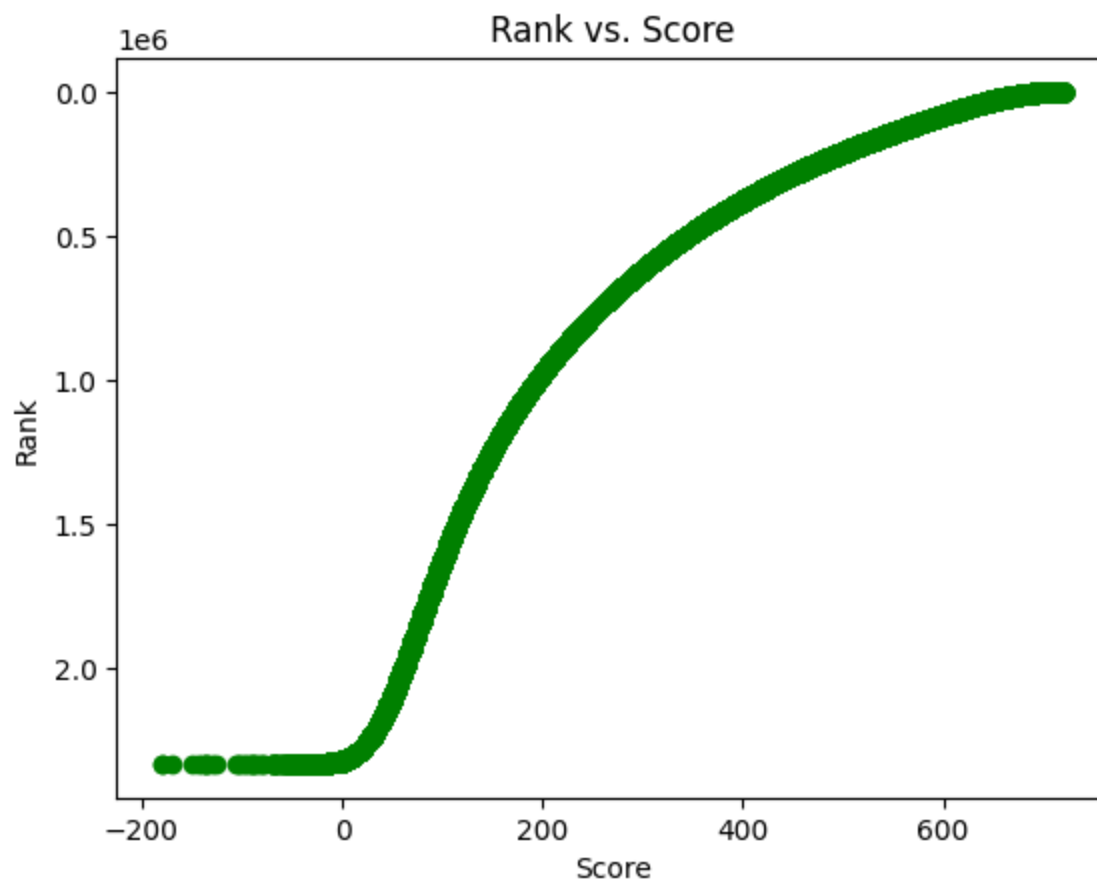
	center_id	center_name	center_city	center_state	serial_no	score	Rank	AIIMS Delhi	AIIMS Bhopal	AIIMS Rishikesh	AIIMS Bhubaneshwai
950	444603	SETH CHANNURAM BAJAJ MEMORIAL PUBLIC SCHOOL, N...	MAINPURI	UTTAR PRADESH	149	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
951	122004	SUBHANIKETAN SCHOOL, NEAR JEWEL MEADOW APARTME...	KAKINADA	ANDHRA PRADESH	171	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
952	120514	SRI VIJNAN VIHARA EM SCHOOL, BESIDE DECOTHLON ...	VIJAYAWADA	ANDHRA PRADESH	294	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
953	390505	LAWRENCE AND MAYO PUBLIC SCHOOL, SECTOR A SRIN...	KOTA	RAJASTHAN	588	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
954	420802	ZEE HIGH SCHOOL HAYATHNAGAR, LANE BESIDE CANAR...	HAYATHNAGAR	TELANGANA	127	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
955	120507	VIKAS GROUP OF INSTITUTIONS, LOYABATA ROAD NUN...	VIJAYAWADA	ANDHRA PRADESH	301	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
956	410609	OM SADHANA CENTRAL SCHOOL, T.P.M. NAGAR,THENI ...	MADURAI	TAMIL NADU	314	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible

	center_id	center_name	center_city	center_state	serial_no	score	Rank	AIIMS Delhi	AIIMS Bhopal	AIIMS Rishikesh	AIIMS Bhubaneswai
957	410113	SAN ACADEMY AGS COLONY PALLIKARANAI CHENNAI TN...	CHENNAI	TAMIL NADU	324	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
958	392321	BHARTIYA MAHILA P.G. MAHAVIDYALAYA, NEAR SANWA...	SIKAR	RAJASTHAN	89	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
959	390551	SOPHIA SR. SEC. SCHOOL, NEW RLY. COLONY, KOTA,...	KOTA	RAJASTHAN	228	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
960	380103	DELHI PUBLIC SCHOOL G T ROAD AMRITSAR PUNJAB, ...	AMRITSAR	PUNJAB	286	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
961	230803	HAPPY SENIOR SECONDARY SCHOOL, NEAR RANI TALAB...	JIND	HARYANA	99	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
962	421907	ACADEMIC HEIGHTS PUBLIC SCHOOL, SURVEY NO. 99,...	MEDCHAL	TELANGANA	9	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
963	380701	LALA SARAN DASS BOOTA RAM AGGARWAL SARV HITKAR...	FAZILKA	PUNJAB	557	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible
964	391804	SETH G B PODAR COLLEGE NAWALGARH, SETH G B POD...	JHUNJHUNU	RAJASTHAN	69	705	833	Not Eligible	Not Eligible	Not Eligible	Not Eligible

Rank vs Score plot

```
In [24]: import matplotlib.pyplot as plt

plt.plot(dataset["score"], dataset["Rank"], marker='o', linestyle='--', color='green')
plt.title("Rank vs. Score")
plt.xlabel("Score")
plt.ylabel("Rank")
plt.gca().invert_yaxis()
plt.show()
```

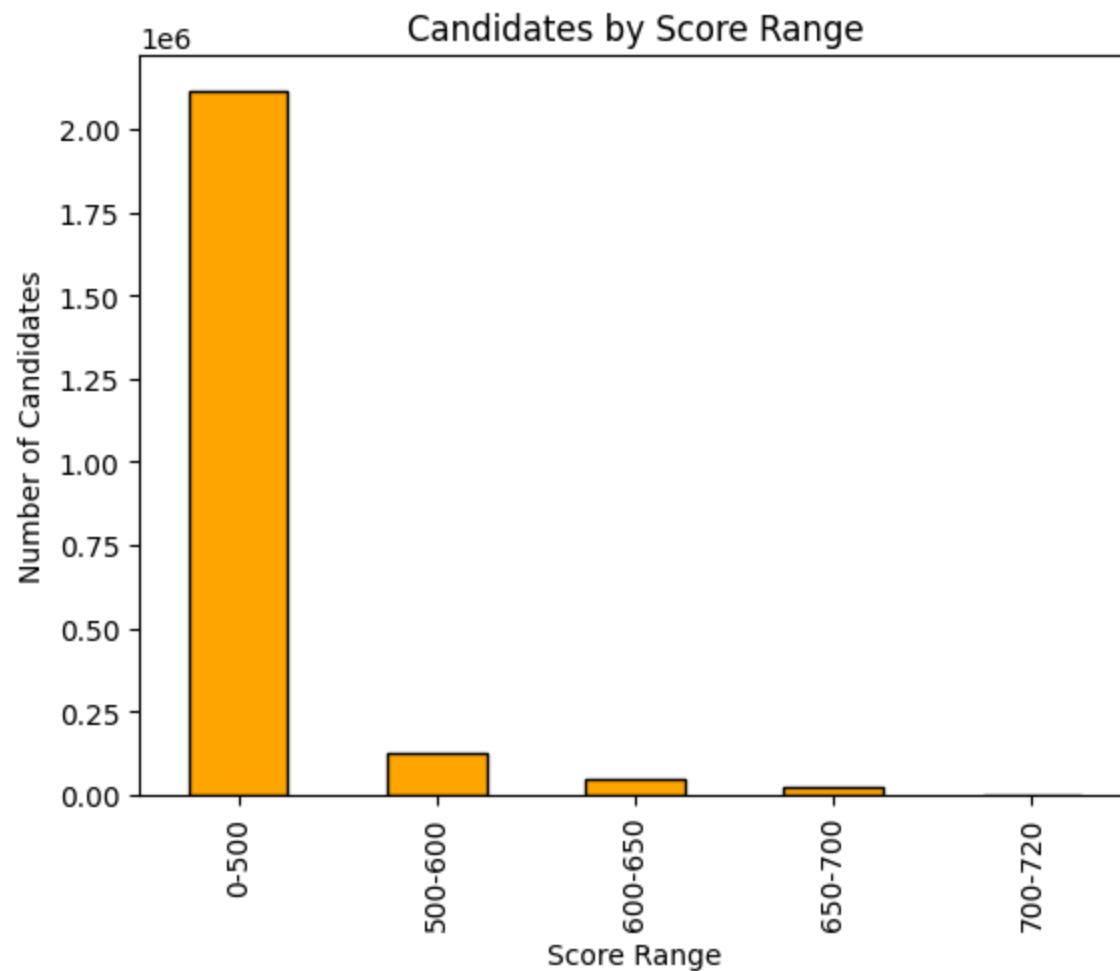


Score Range

```
In [27]: bins = [0, 500, 600, 650, 700, 720]
labels = ['0-500', '500-600', '600-650', '650-700', '700-720']
dataset['Score Range'] = pd.cut(dataset['score'], bins=bins, labels=labels)

score_range_counts = dataset['Score Range'].value_counts(sort=False)

score_range_counts.plot(kind='bar', color='orange', edgecolor='black')
plt.title("Candidates by Score Range")
plt.xlabel("Score Range")
plt.ylabel("Number of Candidates")
plt.show()
```

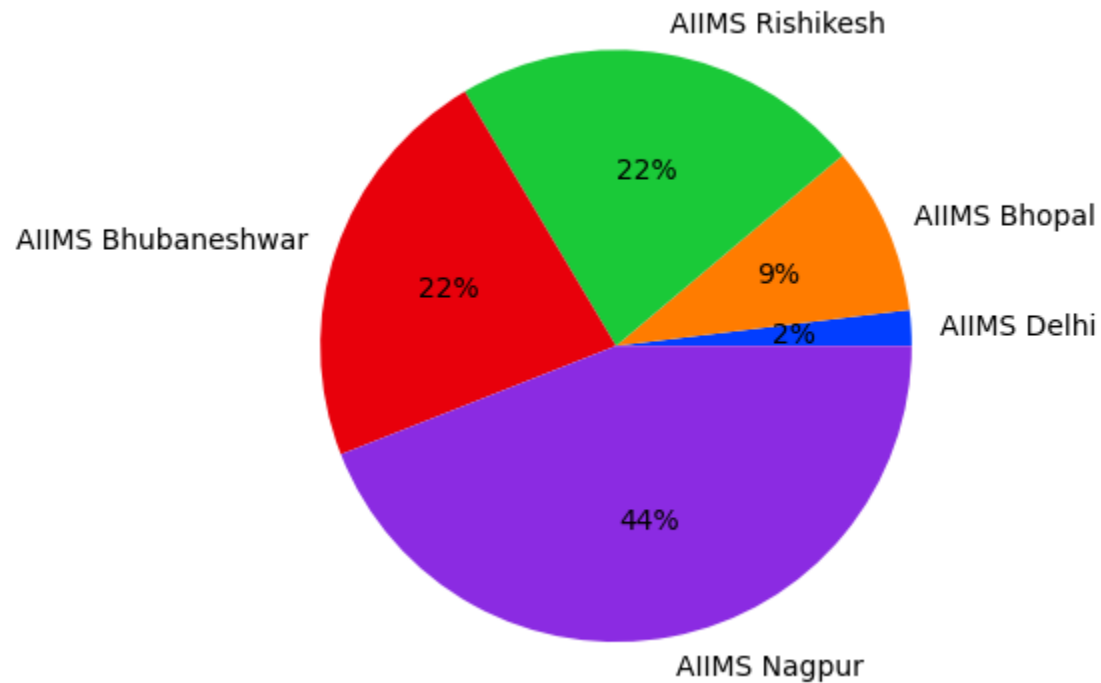


Distribution of candidates eligible for top 5 aiims

```
In [41]: eligible1 = (dataset["AIIMS Delhi"] == "Eligible").sum()
eligible2 = (dataset["AIIMS Bhopal"] == "Eligible").sum()
eligible3 = (dataset["AIIMS Rishikesh"] == "Eligible").sum()
eligible4 = (dataset["AIIMS Bhubaneshwar"] == "Eligible").sum()
eligible5 = (dataset["AIIMS Nagpur"] == "Eligible").sum()

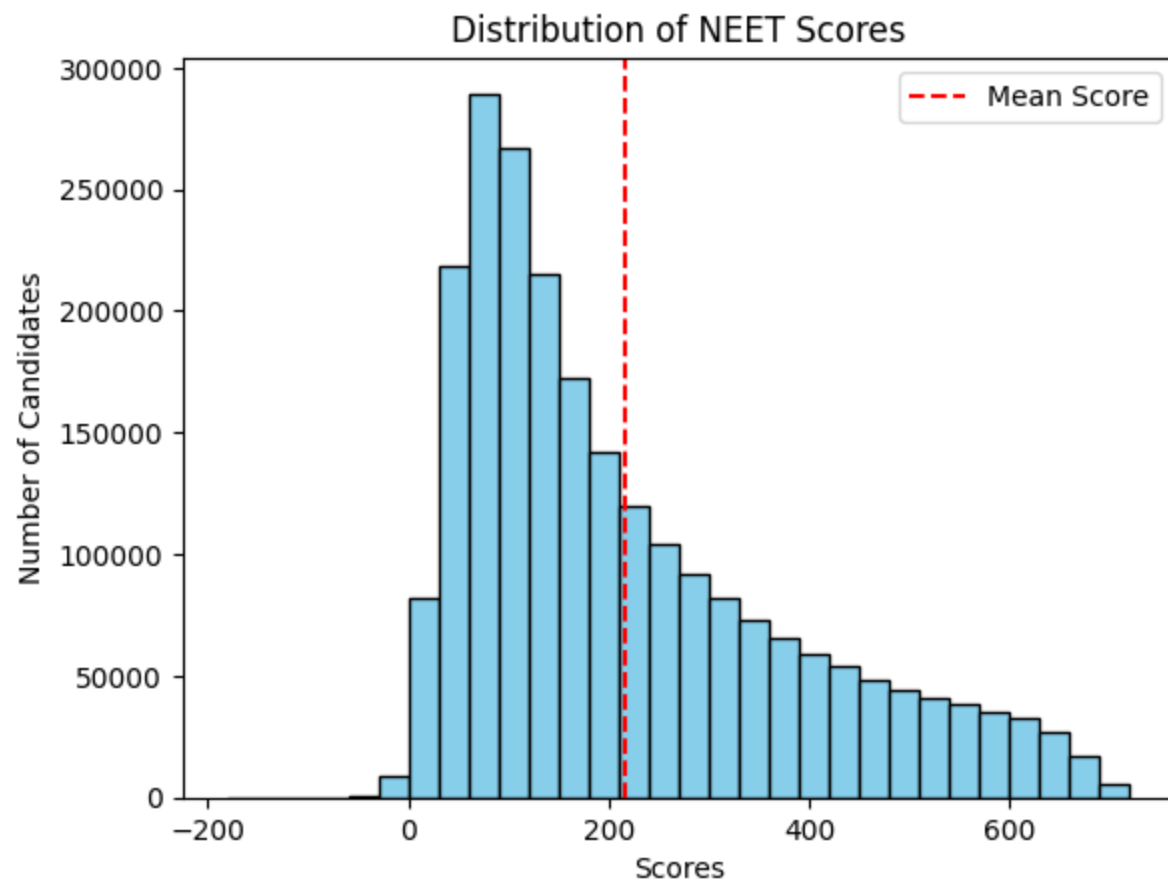
import seaborn
palette_color = seaborn.color_palette('bright')
```

```
pieData = [eligible1, eligible2, eligible3, eligible4, eligible5]
keys = ["AIIMS Delhi", "AIIMS Bhopal", "AIIMS Rishikesh", "AIIMS Bhubaneswar", "AIIMS Nagpur"]
plt.pie(pieData, labels=keys, colors=palette_color, autopct='%0f%')
plt.show()
```



Histogram to distribute scores

```
In [46]: plt.hist(dataset["score"], bins=30, color='skyblue', edgecolor='black')
plt.title("Distribution of NEET Scores")
plt.xlabel("Scores")
plt.ylabel("Number of Candidates")
plt.axvline(dataset["score"].mean(), color='red', linestyle='dashed', label='Mean Score')
plt.legend()
plt.show()
```



Heatmap

```
In [48]: import seaborn as sns
heatmap_data = pd.DataFrame({
    "Eligible": [
        (dataset["AIIMS Delhi"] == "Eligible").sum(),
        (dataset["AIIMS Bhopal"] == "Eligible").sum(),
        (dataset["AIIMS Rishikesh"] == "Eligible").sum(),
        (dataset["AIIMS Bhubaneshwar"] == "Eligible").sum(),
        (dataset["AIIMS Nagpur"] == "Eligible").sum()
    ],
    "Not Eligible": [
        (dataset["AIIMS Delhi"] == "Not Eligible").sum(),
```

```
(dataset["AIIMS Bhopal"] == "Not Eligible").sum(),
(dataset["AIIMS Rishikesh"] == "Not Eligible").sum(),
(dataset["AIIMS Bhubaneswar"] == "Not Eligible").sum(),
(dataset["AIIMS Nagpur"] == "Not Eligible").sum()
]
}, index=["AIIMS Delhi", "AIIMS Bhopal", "AIIMS Rishikesh", "AIIMS Bhubaneswar", "AIIMS Nagpur"])

# Plot heatmap
plt.figure(figsize=(10, 6))
sns.heatmap(heatmap_data, annot=True, cmap="Blues", linewidths=0.5, fmt="d")
plt.title("Eligibility Counts Across AIIMS Campuses")
plt.xlabel("Eligibility Status")
plt.ylabel("AIIMS Campus")
plt.show()
```

